NETWORKED CONTROL
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EXERCIZES

1) Consider the following continuous-time model of system $\mathcal{S}$.

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 4 \\ 4 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

A) Answer to the following questions.

a. Compute the eigenvalues and the modes of the model. Is the system open-loop asymptotically stable?
b. Does the model display unobservable/uncontrollable modes? In case of a positive answer, indicate it/them.
c. Assume that the system is controlled using a <u>continuous-time control law</u> of the type $u(t) = Ky(t)$. Define the range of values of K that make the closed-loop system asymptotically stable.
d. Verify that the model above is equivalent to the following one
$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$
and define the minimal model (i.e., the one whose mode/modes are observable and controllable at the same time) of the system $\mathcal{S}$.

B) We aim to control the system $\mathcal{S}$ using a digital quantized controller. Answer the following questions.

a. According to the theory, what is the minimum rate N_B/h (where N_B is the number of bits encoding the control law and h is the sampling time) required to make the system boundable?
b. In view of the answer given to the previous question, what is the maximum value of h that can be used to make the system boundable in case $N_B = 2$?
c. Set $h = \log_e \sqrt{2}$ s and $N_B = 2$. To control the system, we use the control law defined in case we operate at the rate limits (**with** $u_{\min} = 0$ **and** $u_{\max} = 2$) applied to the observable state of the system. Analyze the trajectories of $y(t)$ using the graphical method (possibly starting from different selected initial conditions). According to the results of the graphical method and to the theory, what is/are the corresponding invariant set/sets where $y(t)$ is bounded to lie asymptotically?
d. **Under the control law defined at point c, define the qualitative asymptotic behavior of the overall state of the original model of the system $\mathcal{S}$. Adequately justify the answer**.

2) Answer the following questions.

a. What is the meaning of the expression "to operate at the rate limit" in the context of control under data rate limitations and quantization? What are the limitations imposed when we operate at the rate limit?
b. What are the opportunities when we operate above the rate limit?
c. Can we operate below the rate limit? Justify the response.

3) Assume that a linear system $\mathcal{S}$ is controlled through the networked control system (NCS) sketched in the following figure.

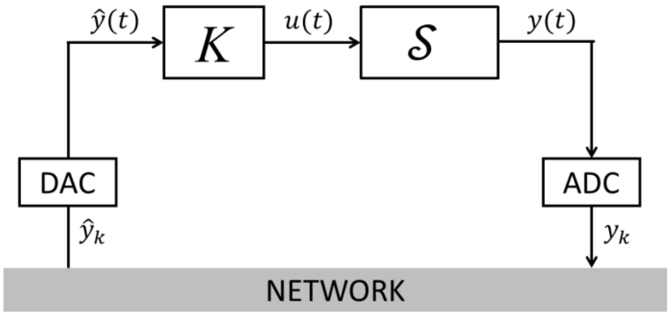

Figure 1: Networked control system

Define the models used to describe the dynamics of the NCS in the following cases:

a. **Constant** sampling time and delay; **no dropouts**.
b. **Time-varying** sampling time and delay; **no dropouts**.
c. **Constant** sampling time and delay; **with dropouts** (with a given deterministic dropout rate r)**.**

In which model classes can the models above be classified? Highlight the main differences between the models defined above and discuss, in a concise way, on how can stability be assessed in these three cases.